Transit Through the L_1 Neck of the Earth–Moon System: Lyapunov Orbits and Their Invariant Manifolds in the Circular Restricted Three-Body Problem

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The planar circular restricted three-body problem of the Earth–Moon system is implemented in MATLAB and used to explain why a spacecraft with enough energy to pass from the Earth region to the Moon region often fails to do so. The equations of motion, the Jacobi integral and the zero-velocity curves are used to find the energy range in which only the neck around the L_1 libration point is open. A numerical trial at a Jacobi constant of 3.18 shows that only launch directions within a window of about 63° lead to a fast transit. The linearized dynamics around L_1 give a saddle–center structure, from which planar Lyapunov orbits are computed by differential correction with the state transition matrix and extended to the target energy by natural-parameter continuation. The resulting orbit has an x-amplitude of about $4\,100$ km, a period of 11.8 days and a monodromy matrix with a real eigenvalue of about $2\,500$. Its stable and unstable invariant manifolds form tubes connecting the Earth and Moon regions. Intersecting the stable tube with the launch line predicts a transit window from -5.3° to 58.1°, in agreement with a direct scan of launch directions within 1°.

Nomenclature

A	=	Jacobian of the equations of motion, 6×6
A_x	=	x-amplitude of a Lyapunov orbit, measured from L_1 towards the Earth
C	=	Jacobi constant
c_2	=	coefficient of the linearized potential at a collinear point
L_i	=	libration (Lagrange) points, $i = 1, \ldots, 5$
M	=	monodromy matrix, $\Phi(T, 0)$
r_1, r_2	=	distances to the Earth and to the Moon
T	=	period of a periodic orbit
t	=	non-dimensional time
U	=	effective potential of the rotating frame
V	=	speed in the rotating frame
V_0	=	launch speed
$\boldsymbol{X}$	=	state vector, $(x, y, z, \dot{x}, \dot{y}, \dot{z})$
x, y, z	=	coordinates in the rotating frame, origin at the barycenter
Y^s, Y^u	=	stable and unstable eigenvectors of M
γ_i	=	distance from L_i to the nearer primary
ϵ	=	displacement used to start the manifolds
θ	=	launch direction, measured from the $+x$ axis
λ	=	real eigenvalue of the linearized in-plane motion at L_1
λ_1, λ_2	=	real eigenvalues of M, $\lambda_1 \lambda_2 = 1$
μ	=	mass parameter, $m_2/(m_1 + m_2)$
ν, ν_z	=	in-plane and out-of-plane frequencies at L_1
τ	=	amplitude ratio of the linear Lyapunov orbit
Φ	=	state transition matrix

I. Introduction

The circular restricted three-body problem (CR3BP) describes the motion of a spacecraft of negligible mass under the effect of the gravity of two primaries that move on circular orbits about their common center of mass. In the Earth–Moon system it is the simplest model capable of capturing the libration points and the low-energy transfers used by missions to cislunar space. Its only integral of motion, the Jacobi constant, divides space into regions the spacecraft can and cannot reach, but it says nothing about whether a trajectory that is allowed to reach the Moon will actually do so.

This paper follows that question step by step. Section II introduces the model, the Jacobi integral and the zero-velocity curves, and Section III locates the libration points. Section IV presents a numerical experiment: at an energy at which only the L_1 neck is open, spacecraft are launched from a fixed point in all directions, and only a narrow window of directions transits. Sections V and VI build the tools needed to explain the window: the linear dynamics near L_1, the state transition matrix, and the computation of planar Lyapunov orbits by differential correction and continuation. Section VII computes the invariant manifolds of the Lyapunov orbit and shows that the stable manifold tube predicts the transit window.

The formulation follows the course reader of Wakker [1] and the book of Koon, Lo, Marsden and Ross [2], both used as guides in the Fundamentals of Astrodynamics course at Delft University of Technology. All computations were carried out in MATLAB; the code is available at `https://github.com/alvaro-sanroman/cr3bp-earth-moon`.

II. The Circular Restricted Three-Body Problem

A. Equations of Motion

The Earth (m_1) and the Moon (m_2) move in circles about their barycenter. In a frame of reference that rotates with them, with the barycenter at the origin and the x axis through both primaries, and in units in which the Earth–Moon distance, the sum of the masses and the angular velocity of the frame are all equal to one, the Earth is at $(-\mu, 0, 0)$ and the Moon at $(1-\mu, 0, 0)$. The equations of motion of the spacecraft are [1, 2]

$$\ddot{x} - 2\dot{y} = \frac{\partial U}{\partial x}, \qquad \ddot{y} + 2\dot{x} = \frac{\partial U}{\partial y}, \qquad \ddot{z} = \frac{\partial U}{\partial z}, \tag{1}$$

with the effective potential

$$U = \frac{1}{2}\left(x^2 + y^2\right) + \frac{1-\mu}{r_1} + \frac{\mu}{r_2}, \qquad r_1 = \sqrt{(x+\mu)^2 + y^2 + z^2}, \qquad r_2 = \sqrt{(x-1+\mu)^2 + y^2 + z^2}. \tag{2}$$

The terms $2\dot{y}$ and $2\dot{x}$ are the Coriolis accelerations and the quadratic term of U is the centrifugal potential. Table 1 lists the constants used. The equations are integrated in first-order form with `ode113`, the variable-order Adams–Bashforth–Moulton solver of MATLAB, with relative and absolute tolerances of 10^{-12}, while the manifold tubes are propagated with a relative tolerance of 10^{-11}.

Table 1 Earth–Moon constants and units

Quantity	Value	Note
Mass ratio m_1/m_2	81.30056	
Mass parameter μ	0.0121505856	$1/(1 + 81.30056)$
Unit of length	384 400 km	Earth–Moon distance
Unit of time	4.3484 days	sidereal month / 2π
Unit of velocity	1.0232 km/s	

B. Jacobi Integral and Zero-Velocity Curves

Multiplying Eqs. (1) by $\dot{x}$, $\dot{y}$ and $\dot{z}$ and adding gives an exact integral, the Jacobi constant [1]

$$C = 2U(x, y, z) - V^2, \qquad V^2 = \dot{x}^2 + \dot{y}^2 + \dot{z}^2. \tag{3}$$

A larger C means a lower energy. Since $V^2 \geq 0$, a spacecraft with a given C can only be where $2U \geq C$. The boundary $2U = C$ is the zero-velocity curve (in the plane) or surface (in space), and the region $2U < C$ is forbidden. The drift of C is also a direct measure of the integration error: for a trajectory starting at $(x, y) = (0.8, 0)$ with $\dot{y} = 0.3$, integrated for 130 days, C stays at 3.112041 with a maximum deviation of 2.9×10^{-11}.

Figure 1 shows the forbidden region for decreasing values of C. For $C > C_{L_1}$ the spacecraft is trapped either around the Earth or around the Moon. When C falls below C_{L_1}, a neck opens around L_1 and the Earth and Moon regions are connected. Below C_{L_2} a second neck opens at L_2 towards the exterior region. The rest of the paper uses $C = 3.18$, for which $C_{L_2} < C < C_{L_1}$: the only way between the Earth and the Moon regions is the L_1 neck, and the Moon region is closed on the far side at L_2.

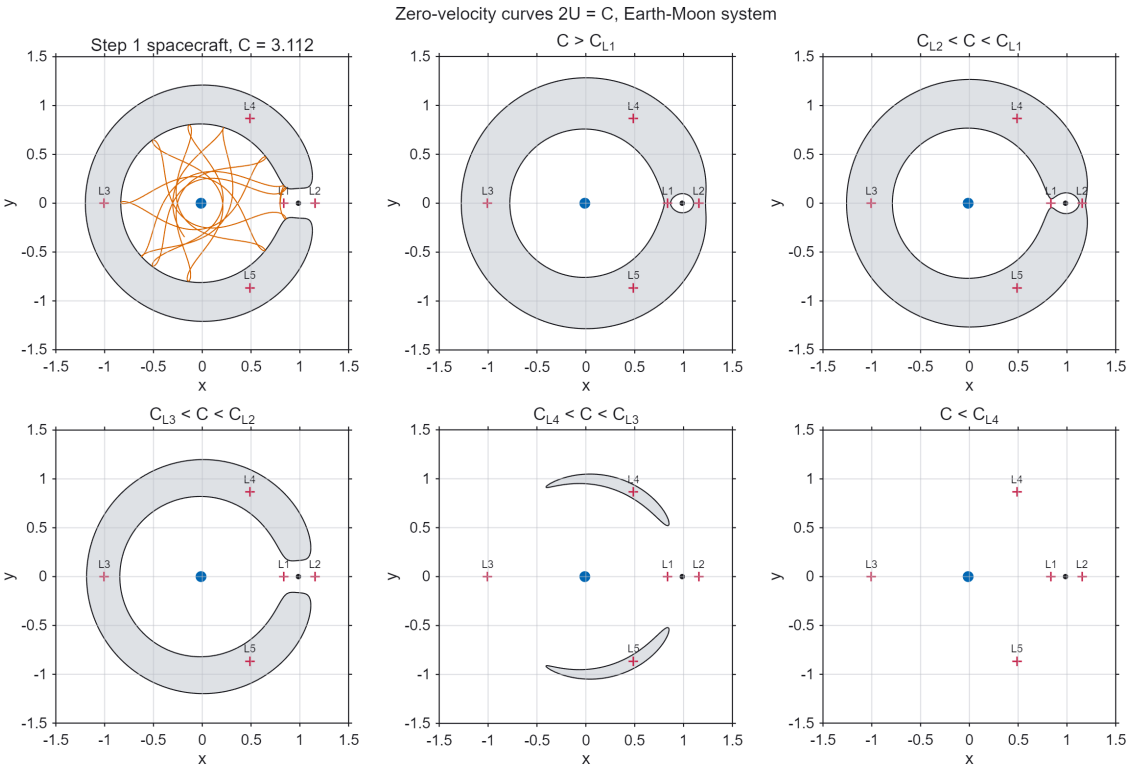


Fig. 1 Zero-velocity curves and forbidden regions (shaded) for decreasing values of the Jacobi constant.

III. Libration Points

At the libration points a spacecraft can stay at rest in the rotating frame, so all the derivatives of U vanish. On the x axis this gives a single equation in x [1],

$$x - \frac{(1-\mu)(x+\mu)}{|x+\mu|^3} - \frac{\mu(x-1+\mu)}{|x-1+\mu|^3} = 0, \tag{4}$$

which has one root in each of the three intervals separated by the primaries. They are found with `fzero`. The equilateral points L_4 and L_5 form equilateral triangles with the primaries. Table 2 lists the results and their Jacobi constants, obtained from Eq. (3) with $V = 0$. The distances γ_i agree with the values tabulated by Wakker [1] to the digits given; the small difference in γ_2 comes from the series expansion used for the table, which differs from the exact root by about 5×10^{-5}. At the computed points the accelerations given by Eqs. (1) are below 4×10^{-16}.

Table 2 Libration points of the Earth–Moon system

Point	x	y	C	γ_i (Ref. [1])
L_1	0.8369151	0	3.188341	0.150934 (0.1509)
L_2	1.1556822	0	3.172160	0.167833 (0.1679)
L_3	−1.0050626	0	3.012147	0.992912 (0.99291)
L_4, L_5	0.4878494	±0.8660254	2.987997	–

L_1 lies about 58 000 km from the Moon and L_2 about 64 500 km beyond it. L_1 is not the point where the attractions of the Earth and the Moon cancel, which is closer to the Moon ($\gamma = 0.0998$, about 38 400 km): a body at rest in the rotating frame moves on a circle in inertial space, so the centrifugal term must also be balanced.

IV. A Numerical Transit Experiment

To see what an open neck means in practice, spacecraft are launched from $(x, y) = (0.80, 0)$, on the Earth side of L_1, with $C = 3.18$. The launch speed follows from Eq. (3),

$$V_0 = \sqrt{2U(0.80, 0) - C} = 0.1485, \tag{5}$$

about 152 m/s in the rotating frame, and only the direction θ is varied, in steps of 5°. All 72 trajectories have the same energy and all could, in principle, reach the Moon region.

Figure 2 shows the result. Within 26 days, only the directions between −5° and 60° cross L_1, and they do so in 1.5 to 3.2 days, passing no closer than about 2 900 km from the center of the Moon; the rest return towards the Earth. This window is not centered on the Moon: the Coriolis term in Eqs. (1) bends trajectories moving in $+x$ towards $-y$, so the spacecraft must aim above the neck. The crossing time grows towards the edges of the window, where trajectories linger near L_1 before falling to one side. In longer integrations a few more directions transit after several loops around the Earth, scattered without a pattern, reflecting the sensitivity of the problem to initial conditions.

The crossing of $x = x_{L_1}$ used in this experiment slightly overestimates the window: the 60° trajectory crosses L_1 but turns back before reaching the Moon. With a stricter criterion, which requires the trajectory to reach $x = 1 - \mu$ before returning to the Earth side ($x = -\mu$),13 of the 14 directions transit. A finer scan of θ in 1° steps with the strict criterion gives transit for $-5° \le \theta \le 58°$, a window of 63°, and no other transits within 26 days.

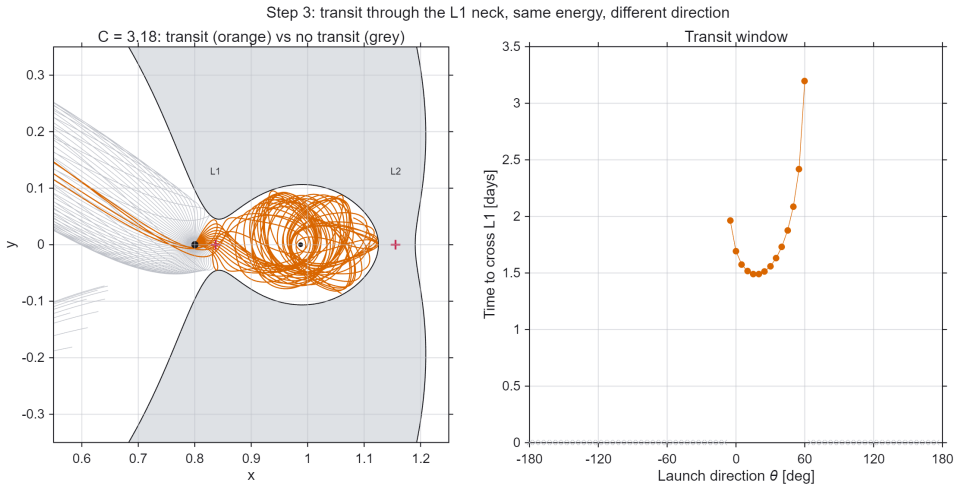


Fig. 2 Transit experiment at $C = 3.18$: trajectories launched from $(0.80, 0)$ in all directions (left) and time to cross L_1 against launch direction (right).

The energy, therefore, only decides what is possible. The rest of the paper explains which trajectories transit, using the dynamics near L_1.

V. Linear Dynamics Near L_1

Expanding Eqs. (1) to first order about a collinear point, with (x, y, z) now measured from the point, gives [2]

$$\ddot{x} - 2\dot{y} - (1 + 2c_2)\, x = 0, \qquad \ddot{y} + 2\dot{x} + (c_2 - 1)\, y = 0, \qquad \ddot{z} + c_2\, z = 0, \tag{6}$$

where $c_2 = (1-\mu)/|x_L + \mu|^3 + \mu/|x_L - 1 + \mu|^3$; at L_1, $c_2 = 5.1476$. The out-of-plane motion is a harmonic oscillation with frequency $\nu_z = \sqrt{c_2} = 2.2688$. The in-plane characteristic equation has one real pair $\pm\lambda$ and one imaginary pair $\pm i\nu$,

$$\lambda^2 = \frac{c_2 - 2 + \sqrt{9c_2^2 - 8c_2}}{2}, \qquad \nu^2 = \frac{2 - c_2 + \sqrt{9c_2^2 - 8c_2}}{2}, \tag{7}$$

which give $\lambda = 2.9321$ and $\nu = 2.3344$. L_1 is therefore a saddle × center × center: unstable, but with oscillatory directions as well. A perturbation along the unstable direction grows by $e^{\lambda} \approx 19$ per time unit, that is, every 4.35 days.

Removing the exponential terms leaves a bounded solution, the linear Lyapunov orbit,

$$x = -A_x \cos \nu t, \qquad y = -A_x\, \tau \sin \nu t, \qquad \tau = -\frac{\nu^2 + 2c_2 + 1}{2\nu} = -3.5865, \tag{8}$$

an ellipse 3.6 times longer in y than in x with a period $2\pi/\nu = 11.704$ days. The four possible combinations of the exponential terms give the four classes of motion near the neck identified by Conley [2]: periodic, asymptotic to the periodic orbit, transit and non-transit. Figure 3 shows them in the McGehee representation, in which the region around L_1 is drawn as a spherical shell. The boundary between transit and non-transit motion is formed by the orbits asymptotic to the Lyapunov orbit.

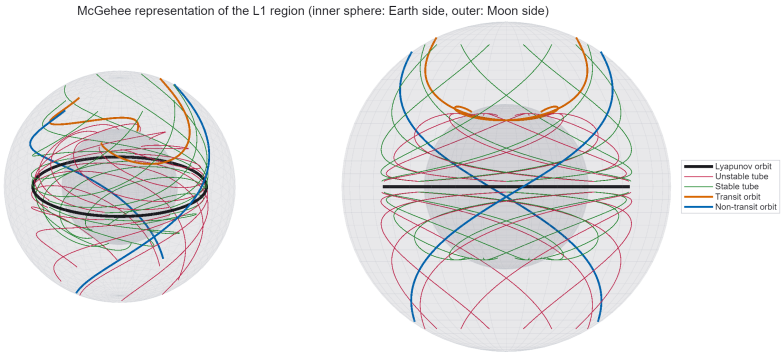


Fig. 3 McGehee representation of the region around L_1 with the Earth–Moon values of λ and ν.

VI. Lyapunov Orbits

A. State Transition Matrix

The state transition matrix $\Phi(t, 0) = \partial \boldsymbol{X}(t)/\partial \boldsymbol{X}(0)$ describes how a small change in the initial state propagates along a trajectory. It is obtained by integrating the variational equations along with the state [2],

$$\dot{\Phi} = A(t)\, \Phi, \qquad \Phi(0, 0) = I_6, \qquad A = \begin{bmatrix} 0 & I_3 \\ U_{XX} & 2\Omega \end{bmatrix}, \qquad \Omega = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \tag{9}$$

where U_{XX} is the matrix of second derivatives of U evaluated on the trajectory. The system has 42 equations: six for the state and 36 for Φ.

B. Differential Correction

The linear orbit of Eq. (8) is not periodic in the nonlinear problem and needs correction. The equations are symmetric under $y \to -y$, $t \to -t$, so an orbit that crosses the x axis perpendicularly twice is periodic. Starting from $\boldsymbol{X}_0 = (x_0, 0, 0, 0, \dot{y}_0, 0)$ with the linear guess $\dot{y}_0 = -A_x \nu \tau$, the trajectory is integrated to its next crossing of $y = 0$, where in general $\dot{x}_1 \neq 0$. Keeping x_0 fixed and accounting for the change in crossing time, the correction to the initial velocity is [2]

$$\delta \dot{y}_0 = \frac{\dot{x}_1}{\Phi_{45} - \Phi_{25}\, \ddot{x}_1 / \dot{y}_1}, \tag{10}$$

which is Newton's method applied to $\dot{x}_1(\dot{y}_0) = 0$. For $A_x = 10^{-3}$ (384 km) it changes the initial velocity by 0.7% and $|\dot{x}_1|$ drops from 7.6×10^{-4} to 4.8×10^{-14} in four iterations, with the quadratic convergence expected of Newton's method (Fig. 4). The corrected period is 11.705 days, against 11.704 days in the linear theory.

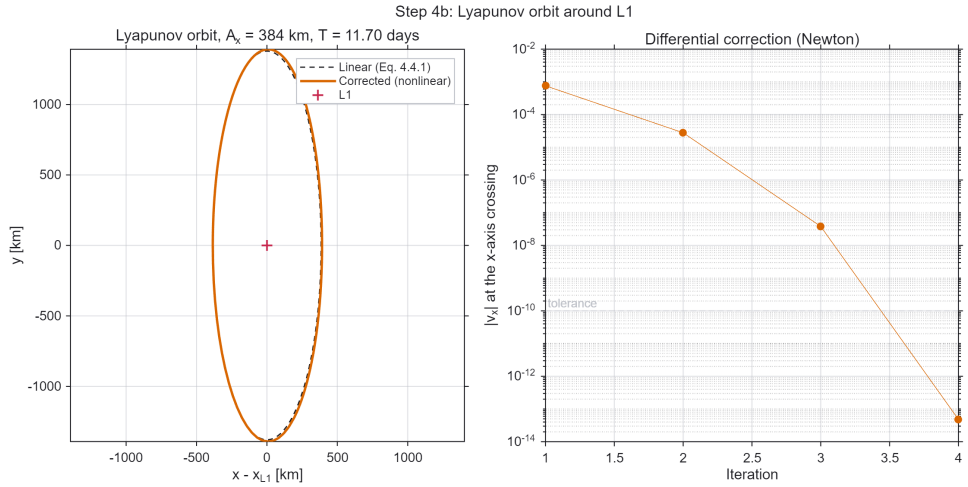


Fig. 4 Corrected Lyapunov orbit compared with the linear ellipse (left) and convergence of the differential correction (right).

C. Monodromy Matrix

The state transition matrix after one period, $M = \Phi(T, 0)$, is the monodromy matrix. Because the problem is Hamiltonian, its eigenvalues appear in reciprocal pairs [2]: a real pair $\lambda_1 > 1$, $\lambda_2 = 1/\lambda_1$, which gives the instability of the orbit; a pair equal to one, associated with the periodic orbit itself and with the family; and a complex pair of modulus one, $e^{\pm i \nu_z T}$, from the out-of-plane oscillation. For the 384 km orbit, $\lambda_1 = 2674$ and $e^{\lambda T} = 2677$, and the complex pair $0.98454 \pm 0.17517i$ matches $\cos \nu_z T \pm i \sin \nu_z T = 0.9846 \pm 0.1751i$ within 2×10^{-4}, so the nonlinear results connect back to the linear theory.

D. Continuation to the Target Energy

As the amplitude grows the linear guess becomes poor. Larger orbits are obtained by natural-parameter continuation [2]: from two corrected orbits, the initial velocity of the next one is extrapolated linearly in A_x and then corrected. Along the family the Jacobi constant decreases from C_{L_1}, and once two orbits bracket $C = 3.18$ the amplitude is refined by the secant method until $|C - 3.18| < 10^{-10}$. Figure 5 shows the family and Table 3 the target orbit.

The orbit is no longer symmetric about L_1: it extends further towards the Moon than towards the Earth, a nonlinear effect absent from Eq. (8). A perturbation along Y^u grows by a factor of almost 2 500 per revolution.

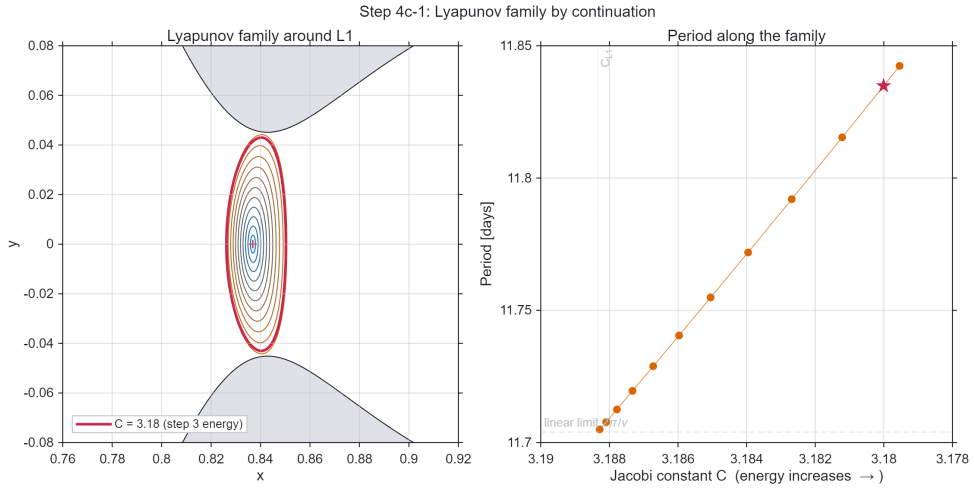


Fig. 5 Family of planar Lyapunov orbits around L_1 obtained by continuation, and the orbit with $C = 3.18$.

Table 3 Lyapunov orbit around L_1 with $C = 3.18$

Quantity	Value
A_x	0.010737 (4 127 km)
Initial state x_0, $\dot{y}_0$	0.8261785, 0.0979481
Extent in x about L_1	−4 127 km to +5 127 km
Extent in y	±16 533 km
Period T	2.721679 (11.835 days)
Eigenvalues λ_1, λ_2	2484.7, 4.025×10^{-4}
Out-of-plane pair	$0.99362 \pm 0.11276i$

VII. Invariant Manifolds and the Transit Window

A. Computing the Tubes

The eigenvectors Y^u and Y^s of M give the directions in which nearby trajectories leave or approach the orbit. Starting from a point $\boldsymbol{X}_0$ on the orbit, the manifolds are initialized by a small displacement [2]

$$\boldsymbol{X}^{u,s} = \boldsymbol{X}_0 \pm \epsilon\, Y^{u,s}(\boldsymbol{X}_0), \qquad \epsilon = 10^{-6}, \tag{11}$$

where the sign selects the branch, towards the Moon or towards the Earth. The unstable manifold is integrated forwards in time and the stable manifold backwards. To start from other points of the orbit, the eigenvectors are transported with the state transition matrix and normalizzed again,

$$Y^{u,s}\big(\boldsymbol{X}(t)\big) = \Phi(t,0)\, Y^{u,s}(\boldsymbol{X}_0). \tag{12}$$

Repeating this for points spread along the orbit gives the tubes of Fig. 6. The Moon branches are integrated up to $x = 1 - \mu$ and the Earth branches up to $x = 0.80$, the launch line of Section IV. Because ϵ is small, trajectories first wind around the orbit for more than one revolution prior to leaving it, and reach these sections after 17 to 19 days.

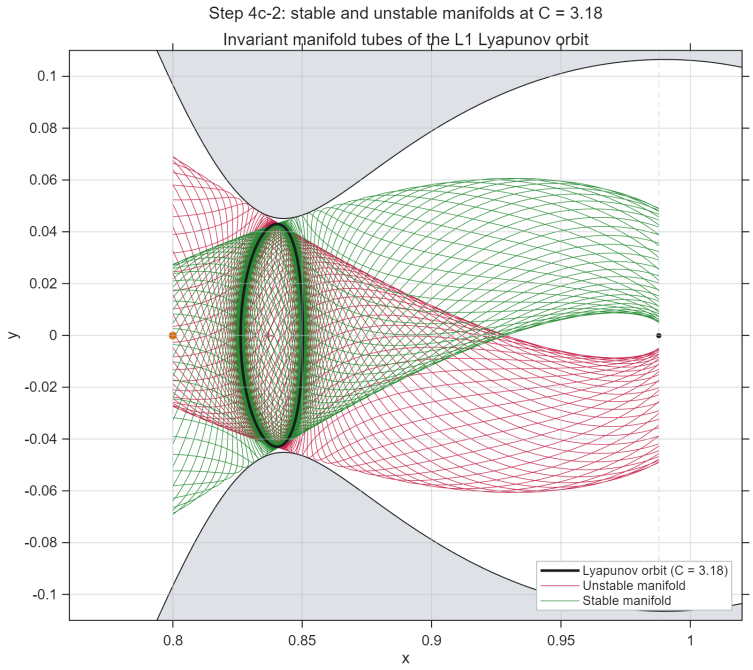


Fig. 6 Stable and unstable manifold tubes of the $C = 3.18$ Lyapunov orbit, Earth branches to $x = 0.80$ and Moon branches to $x = 1 - \mu$.

B. The Tube as a Separatrix

In the planar problem with fixed C, the phase space is three-dimensional, and the stable manifold, two-dimensional, divides it. Trajectories inside the tube transit through the neck; trajectories outside it bounce back. At $x = 0.80$, with $\dot{x} > 0$ fixed by C, the stable tube cuts the $(y, \dot{y})$ plane in a closed curve. The launch states of Section IV lie on the line $y = 0$ with $\dot{y} = V_0 \sin\theta$, so the transit window is the part of that line inside the curve (Fig. 7).

The intersection gives transit for

$$-5.27^\circ < \theta < 58.05^\circ, \tag{13}$$

in agreement with the direct scan ($-5^\circ \le \theta \le 58^\circ$). Applied to the 72 directions of the coarse experiment, the tube classifies all of them as the strict criterion does: the 13 directions from -5° to 55° transit and the rest do not. The 60° trajectory, counted as a transit in Section IV because it crossed L_1, lies just outside the tube, which explains why it turns back. A trajectory launched at $\theta = 28^\circ$, inside the window, crosses L_1 after 1.54 days, passes $x = 1 - \mu$ after 4.78 days, comes within about 6 300 km of the center of the Moon and then stays in the Moon region, since at this energy the L_2 neck is closed.

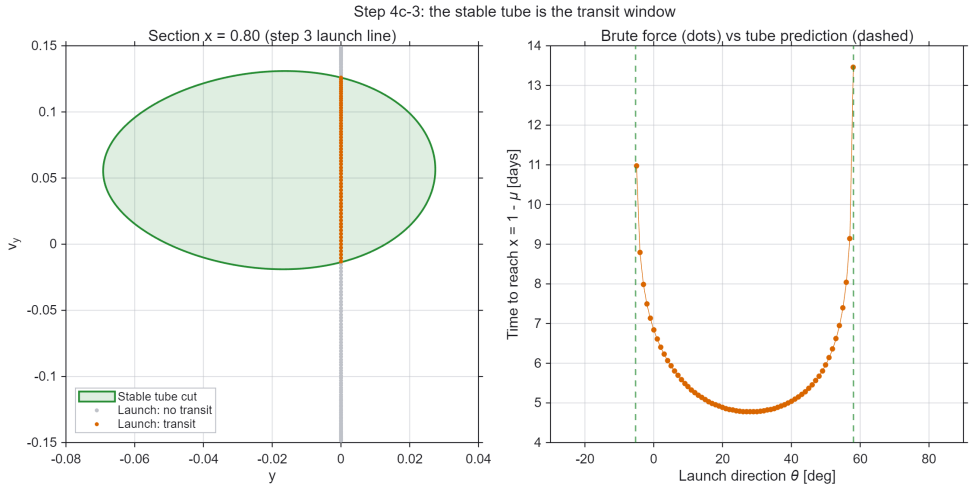


Fig. 7 Cut of the stable tube at $x = 0.80$ and the launch line $y = 0$: the intersection predicts the transit window.

VIII. Conclusions

The planar Earth–Moon CR3BP has been implemented step by step and each stage has been checked against an independent reference: the conservation of the Jacobi constant, the tabulated libration points, the linear theory and the coarse numerical experiment. The main conclusions are the following:

1) At $C = 3.18$ only the L_1 neck is open, but only about one sixth of the launch directions from $x = 0.80$ transit quickly to the Moon region. The Jacobi constant decides what is possible, not what happens.
2) L_1 has a saddle × center × center structure. The center part gives a family of planar Lyapunov orbits, computed by differential correction with the state transition matrix and continued to the target energy.
3) The monodromy eigenvalues of small orbits match the linear theory, and the orbit at $C = 3.18$ has an unstable eigenvalue of about 2 500.
4) The stable manifold tube of that orbit separates transit from non-transit trajectories: its cut at the launch line predicts the transit window to within the 1° resolution of a direct scan, classifies all 72 directions of the coarse experiment correctly, and explains why the 60° trajectory, which crosses L_1, turns back.

The work is limited to the planar problem. The next step is the spatial problem: halo orbits around L_1 and L_2 and their three-dimensional manifolds.

Acknowledgments

The theory was studied from the course reader of Wakker [1] and the book of Koon, Lo, Marsden and Ross [2], both used as guides in the Fundamentals of Astrodynamics course at Delft University of Technology, which I followed as an Erasmus exchange student.

References

[1] Wakker, K. F., *Fundamentals of Astrodynamics*, Institutional Repository, Delft University of Technology, Delft, 2015, Chap. 3. Course reader of Fundamentals of Astrodynamics, TU Delft.

[2] Koon, W. S., Lo, M. W., Marsden, J. E., and Ross, S. D., *Dynamical Systems, the Three-Body Problem and Space Mission Design*, 2nd ed., Shane D. Ross, Blacksburg, VA, 2022, Chaps. 2 and 4.